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Kappa opioids modulate hedgehog signaling to attenuate osteoarthritis

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Abstract

Objective—Inhibition of Hedgehog (Hh) signaling prevents cartilage degeneration and promotes repair in animal models of osteoarthritis (OA). Current study was designed to explore if modulation of kappa opioid receptor (KOR) signaling may have therapeutic potential for achieving disease-modifying activity in OA models through Hh modulation.

Methods—Normal human primary tissues were obtained from the NDRI. Human OA samples were collected from patients with osteoarthritis undergoing total joint replacements. For *in vivo* studies, a partial medial meniscectomy (PMM) rat model was used. A novel automated 3D indentation tester (Mach-1) was used to quantify articular cartilage thickness and stiffness properties.

Results—We have demonstrated that activation of KOR by a selective peptide agonist JT09 inhibits Hh signaling ($p=0.002$) in normal and osteoarthritic human articular chondrocytes via cAMP/CREB pathway. JT09 markedly decreased matrix degeneration in articular chondrocytes and cartilage explants induced by a Hh agonist, SAG ($p=0.026$). Moreover, *in vivo* application of

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Author Contributions

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JT09 via intra-articular injection significantly (60% improvement, $p=0.021$ for the tibial plateau) attenuated articular cartilage degeneration after PMM. In JT09 treated rats, cartilage content, structure and functional properties were largely maintained, and osteophyte formation was reduced (by 70%, $p=0.005$).

Conclusion—Current study defines a novel mechanism for the role of the KOR in articular cartilage homeostasis and disease, providing a potential unifying mechanistic basis for the overlap in processes and pathologies shown to involve opioid and Hh signaling. Moreover, this study identifies a potential novel therapeutic strategy in which KOR modulation can improve outcomes in patients with arthritis.

Introduction

The Hedgehog (Hh) signaling pathway is a highly conserved cell signaling pathway that plays a key function in normal embryonic development. During long bone ontogenesis, canonical Hh signaling pathways cause growth plate chondrocytes to proliferate and undergo hypertrophy before cellular apoptosis and subsequent mineralization to form bone. At the end of long bones, Hh-programmed mesenchymal cell differentiation and death specifies the location of the joint (1).

More recently, it has been discovered that the role of Hh in development is recapitulated in multiple skeletal and joint system repair and regenerative processes and that Hh dysregulation leads to the development of disease (2). In OA, articular chondrocytes undergo phenotypic and gene expression changes similar to that of hypertrophic growth plate chondrocytes in long bone formation. Reminiscent of its effects in embryogenesis, increased Hh signaling has been shown to drive these changes in articular chondrocytes in adult life, and inhibition of Hh signaling prevents cartilage degeneration and promotes repair in animal models of OA (3).

Joint development and homeostasis are dependent on multiple pathways. In our previous studies, we have shown that the kappa opioid receptor (KOR) is highly upregulated during human development (4). Consequently, we have examined genes that are enriched in human cartilage-committed mesenchymal cells (pre-chondrocytes). Unexpectedly, the gene encoding the KOR, *OPRK1*, showed the highest enrichment relative to other genes, including classic genes expressed in the articular cartilage lineage such as *SOX9* and *COL2A1* (5). We have also shown that KOR knockout mice suffer accelerated cartilage degeneration when compared to WT mice following injury (4). Conversely, *in vitro* activation of KOR signaling, previously thought to provide only pain relief (6), was shown by our group to have direct anabolic effects in cartilage cells. These preliminary studies laid the groundwork for establishing KOR signaling as a potential therapeutic target for not only pain relief, but also prevention of joint cartilage degeneration. However, the precise molecular mechanisms involved in KOR regulation of joint development and health remain largely unsolved, thus limiting conclusions regarding a potential role of the KOR in joint health and disease.

All opioid receptor subtypes, including the KOR, are structurally similar members of the GPCR superfamily, which activate either the $G_{\alpha s}$ or G_i subunits (7). We have previously

demonstrated that prolonged KOR activation promotes increases in the secondary messengers, cAMP and PKA, in articular chondrocytes, indicating upregulation of the G_{α_s} subunit (4). As Hh expression has been shown to be directly inhibited by G_{α_s} -PKA signaling (8), we hypothesized that opioid receptor signaling may function in joint development and homeostasis through modulation of Hh expression, and that opioid receptor signaling may have therapeutic potential for achieving disease-modifying activity through Hh modulation.

Methods

For all experiments, biological replicates (cells from independent specimens) were employed to generate data. For experiments expected to yield large differences, standard practice of using 3–5 replicates was followed.

Isolation and culture of primary human and porcine cartilage tissue

Only early passages of chondrocytes (passage 0–2) were used for experimentation to avoid de-differentiation and loss of cartilage phenotype (9). Normal human male and female adult articular cartilage and synovium (n=6, 20 to 60 years old) were obtained from the National Disease Research Interchange (NDRI). Human OA samples were obtained from NDRI and Dr. Jay Lieberman (protocol HS-16–00449 approved by the USC IRB Committee). These OA samples were dissected from the tibial and femoral articular cartilage. All patients had 3–4 on the Kellgren–Lawrence grading scale for OA. Three male (60, 67 and 79 years old) and three female (55 and 70 and 81 years old) specimens were used for experimentation. All donated material was anonymous and carried no personal identifiers. Samples were processed following previously established protocols (5). Legs from five-month-old Yucatan minipigs were obtained from S&S Farms (Ramona, California), and cartilage was harvested and digested following previously established protocols. Cartilage explants were harvested using 2 mm punch biopsies (Miltex, Inc., York, PA), and the wet weight of each explant determined prior to experimentation. For cell isolation cartilage tissue was digested as described previously (10, 11).

Cell culture reagents were purchased from Life Technologies, Inc. (Grand Island, New York). JT09 drug was a gift from Dr Thomas A. Dix from Department of Drug Discovery and Biomedical Sciences, College of Pharmacy, Medical University of South Carolina, Charleston, SC. Smoothed Agonist (SAG) and PKA inhibitor (H89 hydrochloride) were purchased from Sigma Aldrich (St. Louis, MO). Human and pig chondrocytes were cultured in DMEM F12 medium containing 10% (vol/vol) fetal bovine serum and 1% Penicillin-Streptomycin (vol/vol) at 37 °C in a humidified atmosphere of 95% air and 5% CO₂.

Western Blot analysis

Treated and non-treated articular chondrocytes were dissolved in RIPA Lysis and Extraction Buffer (Pierce, Rockford, IL) containing protease inhibitors (Pierce) followed by sonication with a 15-second pulse at a power output of 2 using the VirSonic 100 (SP Industries Company, Warminster, PA). Protein concentrations were determined by BCA protein assay (Pierce) and boiled for 5 minutes with Laemmli Sample Buffer (Bio-Rad, Hercules, CA). Proteins were separated on acrylamide gel and analyzed by Western blot using antibodies

purchased from Cell Signaling, anti-GLI1 (Cat #2534) and pCREB (Cat #9198) primary antibodies. Histone 3 (Cat #9715) antibody was used as loading control. Proteins were resolved with SDS-PAGE utilizing 4–15% Mini-PROTEAN TGX Precast Gels and transferred to Trans-Blot Turbo Transfer Packs with a 0.2- μ m pore-size nitrocellulose membrane. The SDS-PAGE running buffer, 4–15% Mini-PROTEAN TGX Precast Gels, Trans-Blot Turbo Transfer Packs with a 0.2- μ m pore-size nitrocellulose membrane were purchased from Biorad (Hercules, CA). Nitrocellulose membranes were blocked in 5% nonfat milk in 0.05% (v/v) Tween 20 (PBST) (Corning, Manassas, VA). Membranes were then incubated with primary antibodies overnight. After washing in PBS containing 0.05% (v/v) Tween 20 (PBST), membranes were incubated with secondary antibodies (Thermo Scientific, Rockford, IL). After washing, development was performed with the Clarity Western ECL Blotting Substrate (Biorad, Hercules, CA) and images were quantified by ImageJ software.

Quantitative Real-Time PCR

Power SYBR Green (Applied Biosystems) RT-PCR amplification and detection was performed using an Applied Biosystems Step One Plus Real-Time PCR machine. The comparative Ct method for relative quantification ($2^{-\Delta\Delta Ct}$) was used to quantitate gene expression. TBP (TATA-box binding protein) was used for gene normalization and expressed relative to a calibrator (sample in each set with lowest expression). Primer sequences are provided in Supplementary Table 1.

Experimental Animal Model and Surgical Procedures

All animal experiments were accomplished in accordance with the National Institutes of Health Guidelines for the Care and Use of Laboratory animals, and were approved by the institutional Animal Care and Use Committee (IACUC) at University of Southern California. Sixteen male Sprague-Dawley Rats (12 weeks old; Jackson Labs) were used in the experimental phase of this study. Rats were housed (2 per cage) in a regulated environment (temperature, 20–22°C; humidity, 50 $\pm$ 5%; night/day cycle, 12 hours) with freedom to move about the cage and access to food and water. Each rat was randomized to treatment with JT09 (n=7), saline (n=6), or sham (n=3) (Supplementary Figure 1). To ensure proper treatment allocation, rats were tagged for identification based on their treatment assignments.

Anesthesia was induced and maintained with 2.5% isoflurane. Following anesthetic induction, all rats were subjected to unilateral partial medial meniscectomy (PMM) surgery on the right knee joint (12). Briefly, each rat was surgically prepped and draped in a sterile fashion. A 1 cm incision was then made from the distal patella to the proximal tibia to expose the patellar tendon. Then the joint cavity was opened through the standard medial parapatellar approach. An ophthalmic surgical blade was used to incise the medial meniscotibial ligament. The meniscus was separated from the tibial plateau and then ~50% of it was excised.

Weekly Injections, Joint Harvesting, and Microscopic Analysis

At weekly intervals following the PMM surgery and initial injection, rats received intra-articular injection of their assigned treatment to the operated knee (Supplementary Figure 1). Following induction of anesthesia, the patella and tibial tuberosity were palpated on the right knee. Once the location of the patellar tendon was determined, a needle syringe was inserted lateral to the tendon. Once insertion through the knee capsule was felt, intra-articular placement was confirmed by cycling the knee through flexion and extension to estimate the position of the needle in respect to the patella and other knee landmarks. 100 μ l of the assigned treatment was then injected, and rats were observed for safe return to consciousness and normal activity. Six weeks after the initial surgery, rats were sacrificed.

Prior to fixation and decalcification, all rat femurs and tibias were separated for micro indentation testing. Femurs and tibias had to be separated to meet the requirements for testing with the Mach-1 microindenter.

Microindentation Testing

For microindentation testing, a novel automated 3D indentation mapping mechanical tester was used (Mach-1, Biomomentum) (13). Once harvested, a cyanoacrylate adhesive was used to glue samples to the testing chambers, with care taken to ensure that the articular cartilage of the femoral condyle or tibial plateau was facing up in the direction of the indenter (Supplementary Figures 2 and 3). Proper load cell calibration was then ensured for each sample, and a mapping toolbox software was used to photograph samples and outline the surface of the articular cartilage. With boundaries delineated for the medial and lateral compartments, a position grid was overlaid on each sub-surface (Supplementary Figure 2). Following biomechanical measurements, the spherical probe indenter was replaced by a needle indenter which measures load as it descends first through air, then pierces the articular cartilage, and lastly pierces through the surface of subchondral bone (Supplementary Figure 3). The desired positions for biomechanical or thickness testing were then manually selected. Around 20 positions on each sub-surface were designated, with small variation in position number and placement based primarily on the size of the sample. Additionally, one reference point was selected for each sub-surface. The testing chamber was then filled with PBS solution to cover the highest point of the sample by at least 1 cm, and 15 minutes were allowed to pass for equilibration. The indenter was then lowered to 2 mm above the sample surface, and positioned directly above the reference point for a given sub-surface. The metric coordinates for each reference point was determined, and subsequent measurements of stiffness and thickness were positionally identified on the articular cartilage surface based on their relation to these reference points. To ensure proper determination of the biomechanical properties of the cartilage surface, the spherical probe was used first to measure structural stiffness, prior to the use of a needle probe (which had the potential to alter surface properties after needle penetration) to measure cartilage thickness. Batch analysis software was used to collect all relevant data points following completion of microindentation. A mixed-effects model analysis with Tukey's post hoc for multiple comparisons was conducted to evaluate the differences in articular cartilage thickness and stiffness across different regions using data points collected from multiple

animals (13). Additional detailed information on biomechanical testing is provided in the Supplementary Methods section.

Histology and immunohistochemistry

Rat tissues were fixed in 10% formalin for 48 hours, dehydrated with ethanol, and embedded in paraffin using routine procedures. A microtome (Leica) was used to cut 5- μ m sections. Slides were then deparaffinized and stained for sulfated GAG with Fast Green/Safranin O. Slides were also stained with antibodies pCREB (Cell Signaling, Cat# 2534), GLI1 (Cell Signaling, Cat# 9198), ADAMTS4 (Bioss USA, Cat# bs-1459R), ADAMTS5 (Bioss USA, Cat# bs-3573R), MMP13 (Abgent, Cat# 13706C), COL10 (Abcam, Cat# ab58632) and Kappa Opioid Receptor (GeneTex, Cat# GTX78005). H&E staining was performed to assess morphology.

Histological, Osteophyte Production, and Immunohistochemical Analysis

Medial femoral condyles sections were taken from the medial tibial plateau in sagittal slices at a thickness of 5 microns using a microtome (Leica). Tissue sections from each sample were stained with Safranin O and Fast Green for assessment of proteoglycan distribution and osteophyte production respectively. Samples were then blinded for treatment status and graded according to the Osteoarthritis Research Society International (OARSI) scoring system guidelines (14). Qualitative assessment of cartilage cap continuity and presence of subchondral bone fibrosis and cystic formation were also assessed microscopically.

Neopeptide analysis

Pig articular cartilage explants were cultured with SAG (3 nM), JT09 (10 μ M) or both and then digested for 2 hr at 37°C with 0.01 units chondroitinase ABC (Sigma Aldrich, St. Louis, MO). Samples were then dialyzed with ultrapure water for 24 hr at 4°C, freeze dried, dissolved in RIPA Lysis and Extraction Buffer (Pierce, IL) containing protease inhibitors (Pierce) followed by sonication. Proteins were separated and analyzed by Western blot using primary antibodies against aggrecan (Novus Biologicals, Cat# NB100–74350) and collagen neopeptides (Ibex, Cat #50–1035). Wet weight of explants was used as loading control.

Statistical analysis

Numbers of repeats for each experiment are indicated in the figure legends. Pooled data are represented as mean $\pm$ SD unless otherwise indicated. Unless otherwise indicated, statistical analysis was performed using one-way ANOVA followed by the Tukey test to compare more than 2 groups or 2-tailed Student's *t* test to compare 2 groups. *p* values less than 0.05 were considered to be significant.

Results

JT09 Modulates the Expression of Hedgehog Signaling in Both Healthy and OA Chondrocytes

Our studies have shown that KOR is expressed in developing human chondrocytes (4). To further validate these findings, we examined if this opioid receptor is expressed in adult rat

and human articular chondrocytes. Immunohistochemical analysis has shown that a vast majority of articular chondrocytes are highly positive for KOR (Figure 1A). The *OPRK1* gene was highly expressed in human adult healthy chondrocytes, with no significant differences in the expression levels observed between normal and OA articular chondrocytes (Figure 1B).

Previous studies have shown that inhibition of Hh signaling reduces severity of OA (3, 15). It has also been shown by our group that KOR signaling through cAMP/CREB can impact OA pathogenesis and disease outcomes (4, 5). To determine if Hh and KOR signaling interact to regulate chondrocytes and disease progression, we examined the effects of KOR signaling on the downstream Hh target, GLI1, using a selective KOR agonist, JT09. JT09 is a peptide compound shown in animal models to provide similar pain relief to morphine, but with significantly reduced central availability and addictive potential (6). We first treated healthy pig articular chondrocytes with JT09 for 24 hours at concentrations of 100 nM, 300 nM, 1 μ M and 3 μ M, and measured phosphorylated CREB (pCREB) levels as an indication of PKA activity and GLI1 levels as an indication of Hh activity (Figure 2A). Increasing concentration of JT09 increased pCREB while simultaneously decreasing GLI1 levels in a dose-dependent fashion. Effects of JT09 plateaued at 300 nM, and this concentration was selected for subsequent *in vivo* studies while remaining *in vitro* experiments were conducted using a concentration of 10 μ M.

We have previously shown that KOR null mice show accelerated articular cartilage degeneration following injury relative to WT animals (4). We therefore sought to determine if Hh signaling inhibition through KOR activation could be achieved in human OA chondrocytes. We found that human OA chondrocytes compared to healthy chondrocytes displayed significant decreases and increases in pCREB and GLI1 expression, respectively (Figure 2B), in agreement with previous findings (15). This profile in OA chondrocytes was then abrogated by a 24-hour treatment with JT09, suggesting that JT09 can reduce Hh signaling in OA chondrocytes via increases in intracellular cAMP. Cyclic AMP has been shown to modulate PKA, and PKA activation directly phosphorylates GLI, which represses Hh target gene expression (16, 17). To confirm that JT09 activation of KOR operates through cAMP/CREB to modulate GLI1 expression, we treated pig chondrocytes with a PKA inhibitor, H89 (100 nM, 24 hours), and found that H89 inhibition of PKA/CREB signaling leads to an increase in GLI1 while JT09 decreases GLI1 (Figure 2C). Articular chondrocytes obtained from male and female donors with OA showed similar responses to JT09 treatment (Figure 2B and Supplementary Figure 4).

In previous work with KOR null mice, we found that the null phenotype increased susceptibility to joint damage and more severe cartilage degeneration (4). This suggests that KOR activation regulates Hh signaling following injury, and that this is recapitulated *in vivo*. RT-PCR of human OA chondrocytes with JT09 resulted in a significant decrease in the expression of catabolic genes *MMP13* and *ADAMTS5*, suggesting that KOR modulation of Hh signaling has the propensity to decrease cartilage matrix degradation in the disease progression (Figure 2D). In all, we show that Hh signaling in human OA is upregulated, and that a KOR agonist, JT09, increases pCREB while decreasing GLI1 signaling in the same cells in alignment with our hypothesis. Taken together, these results suggest that activation

of non-canonical KOR signaling through cAMP/CREB can inhibit the downstream Hh target, GLI1, in healthy and OA chondrocytes to promote cartilage protection and repair, and improve disease outcomes.

SAG induces degenerative changes in cartilage cells and matrix that can be directly inhibited by JT09

In order to demonstrate that inhibition of Hh signaling is essential for downstream cartilage protective effects, we have utilized a commonly used activator of Hh signaling, Smoothed Agonist (SAG). This synthetic small molecule is a chlorobenzothiophene-containing compound that acts as an activator of the G protein-coupled receptor Smoothed (18). Smo blockage could be therapeutic in reducing osteoarthritic severity caused by articular cartilage degeneration(19). Cartilage degenerating proteases in articular cartilage such as matrix metalloproteinase 13 (MMP13) and a disintegrin and metalloproteinase with thrombospondin motifs 5 (ADAMTS5) are responsible for the degradation of the major components of this tissue, collagen and aggrecan, respectively. Treatment of pig articular chondrocytes with JT09 ameliorated SAG-induced transcription of MMP13 and ADAMTS5 (Figure 3A). To measure direct collagenase and aggrecanase activity in the matrix, we performed a neo-epitope assay which shows the amount of cleaved collagen and aggrecan neo-epitopes in the supernatant. Treatment with SAG dramatically increased collagenase and aggrecanase activity, and addition of JT09 to SAG-treated explants significantly reduced cartilage degradation (Figure 3B). This indicated that JT09 was able to reverse the effects of SAG, suggesting that Hh inhibition is crucial for prevention of extracellular matrix degeneration.

JT09 Prevents the Progression of Cartilage Damage in Vivo

Since these results suggested that JT09 can inhibit Hh signaling in articular chondrocytes, we next sought to determine if application of JT09 could prevent chondrocyte hypertrophy and cartilage matrix degeneration *in vivo*, a mechanism known to be a key driver of OA (15). To evaluate the cartilage protective effects of JT09, we performed a partial medial meniscectomy (PMM) model of rat knee OA and administered intra-articular injection of saline or 300 nM JT09 in the operative knee at the time of surgery and weekly thereafter for six weeks (Supplementary Figure 1). Osteoarthritis research society international (OARSI) histological scoring and osteophyte size and maturity analysis of Safranin O stained sections indicated dramatic cartilage loss, surface disruption, and osteophyte generation in vehicle rat femur and tibia. In JT09 treated rats, cartilage content and structure were largely maintained, and osteophyte formation was greatly reduced (Figure 4A).

During cartilage degeneration, cells undergo apoptosis, promoting matrix loss via MMPs and aggrecanases leading to structural changes. It has been previously shown that upregulation of Hh signaling increases activation of its transcriptional regulator, GLI1, which leads to dramatically increased expression of genes whose protein products mediate cartilage extracellular matrix destruction including ADAMTS4, ADAMTS5, and MMP13, and markers of chondrocyte hypertrophy like COL10 (15). We next examined the articular cartilage expression of these OA markers in control and JT09-treated rats. IHC revealed dramatically reduced expression of GLI1, ADAMTS4, ADAMTS5, MMP13, and COL10 in

rats treated with JT09 (Figure 4B), in agreement with previous findings demonstrating the reduction of these markers in cartilage by at least half with Hh signaling inhibition (through genetic ablation or pharmacologic agents) (15). In conjunction with our mechanistic results, these data suggest that intra-articular application of a selective KOR agonist, JT09, prevents cartilage matrix degeneration and chondrocyte hypertrophy *in vivo* and, at least partially, this mechanism is likely to be mediated through modulation of Hh signaling. Histological assessment of synovial membrane showed minimal differences between groups; a more prominent mononucleated infiltration in the control group potentially indicating a persistent inflammatory reaction as a result of degenerative changes in articular cartilage (Supplementary Figure 5).

JT09 Improves Articular Cartilage Functional Compressive and Structural Properties

In addition to conventional scoring of cartilage quality, we also used a novel bio-indentation scanner (Mach-1) to determine the biomechanical effect of JT09 on rat knee functional joint health and articular cartilage structural phenotypes across the entire articular surface (Supplementary Figure 2A). Treatment with JT09 significantly improved the mechanical properties of the medial femoral condyle (MFC) and the medial tibial plateau (MTP) when compared to vehicle rats (Figure 5). In particular, qualitative and quantitative results demonstrate that vehicle rats suffered the greatest reduction in mechanical properties in the middle third of the MFC and MTP, in agreement with previous studies demonstrating the greatest risk of cartilage damage in this region following PMM (20) (Supplementary Figure 2B). In contrast, JT09-treated rats demonstrated equivalent mechanical properties in all regions of the MFC and MTP when compared to sham rats, suggesting that treatment with JT09 prevented functional compressive tissue degeneration following PMM (Figure 5).

Vehicle rats demonstrated significant tissue loss and thinning of the articular cartilage surface in the middle third of the MTP and MFC (Figure 5A–D). In the outer and inner thirds, vehicle rats had significantly higher cartilage thickness with substantial variation, suggesting degeneration at different stage. Treatment with JT09 resulted in a grossly smooth and uninterrupted articular surface. Corresponding thickness values demonstrated much less variation, with no significant tissue loss or cartilage thinning evident in the high-risk middle third of the MTP and MFC (Figure 5A,B) compared to sham rats. In agreement with the improved cartilage content, structure, and biochemical profile seen on microscopic analysis, these results indicate that treatment with JT09 prevents degenerative changes to the structural phenotype of the articular surface following PMM.

Discussion

The current study defines a novel molecular mechanism for the role of the KOR in articular cartilage homeostasis and disease, providing a potential unifying mechanistic basis for the overlap in processes and pathologies shown to involve opioid and Hedgehog (Hh) signaling. Moreover, this study identifies a potential novel therapeutic strategy in which KOR modulation in cartilage and/or synovial tissue can improve outcomes in patients with arthritis.

However, this study did include some limitations. For the *in vivo* OA model, the rat sample size was relatively small and they might have not been completely skeletally mature as they were only 12 weeks old. In addition to age, all animals analyzed were males, and gender was not addressed as a biological variable. Although our mechanistic *in vitro* studies clearly established an inhibitory effect of KOR agonists on Hh signaling known to be implicated in the pathogenesis of OA, it cannot be definitively claimed that chondroprotective effects of JT09 *in vivo* is exclusively Hh dependent. More studies are needed to determine the therapeutic potential of KOR agonists in human OA.

As a single morphogen, Hh induces different responses depending on the cell type, Hh signaling dose and time of exposure received by a responding cell (21). This is believed to be the means for the wide breadth of normal, pathologic, and repair/regenerative processes shown to involve Hh signaling (21). Previous studies (22) have indicated that opioids such as morphine may upregulate KOR gene expression; and clinical observation demonstrates losses in bone mineral density and increases in risks of fracture and osteoporosis with chronic prescription opioid use. However, a direct mechanistic link between Hh and opioid signaling has never before been previously made.

For the first time, we have clearly demonstrated the role of opioid receptor activation in modulating Hh signaling, providing a unifying mechanistic basis for the overlap in processes and pathologies shown to involve opioid and Hh signaling. Our results provide a molecular mechanism for the growing body of literature demonstrating the non-pain related peripheral function of the KOR for arthritis prevention (4, 23). However, the therapeutic potential of these findings goes far beyond that, which is demonstrated here in chondrocytes. Modulation of Hh signaling has been identified as a therapeutic target for enhancing regeneration and repair of damaged cartilage tissue. Further, modulation of Hh signaling has important implications for maintenance of skeletal health beyond the joint, including in bone mineralization and bone mass regulation (24). Of significance, the oral peripheral KOR peptide agonist KORSUVA/CR845 is in late-stage FDA clinical trials for its application in providing less addictive pain relief, facilitating the direct translation of these and future findings to repurposed human application. Given the current lack of available treatments for OA and cartilage damage, our findings identify new therapeutic possibilities for KOR agonists. In all, the present study opens up a novel mechanism for the use of opioid signaling to maintain skeletal and joint health and treat conditions in which modulation of Hh signaling can be therapeutically beneficial.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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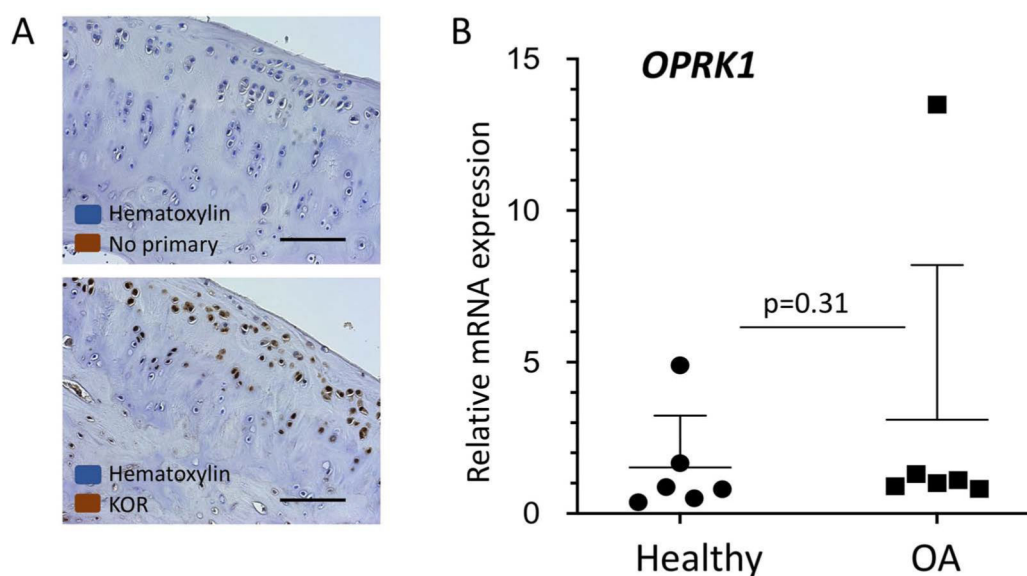


Fig 1. Expression of KOR and *OPRK1* in rat and human cartilage, respectively.

(A) Kappa opioid receptor cross-sectional staining of the articular surface of a 3-month-old rat OA cartilage. Representative images (n=3). Scale bar: 100 μ m. All images taken at 10x magnification. **(B)** *OPRK1* gene (encodes KOR) expression in human healthy and OA cartilage (n=6). P-value calculated using one-way ANOVA with Student's *t* test.

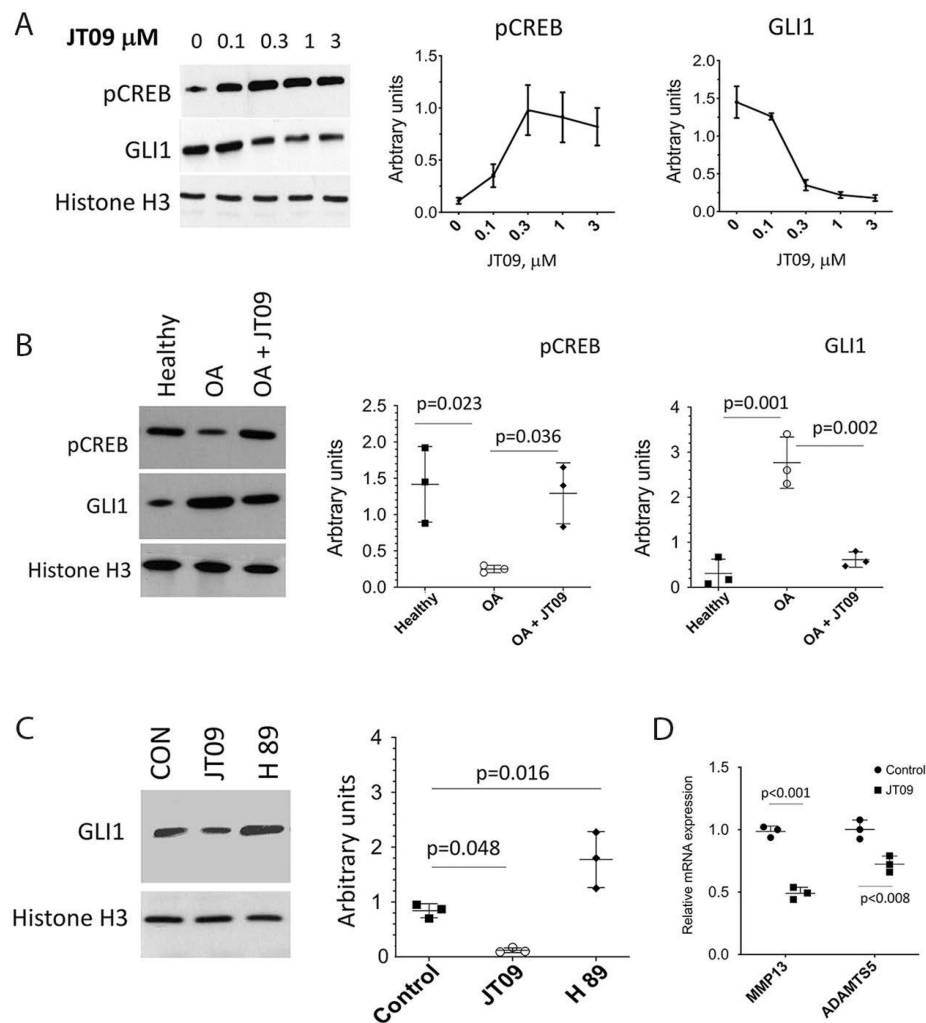


Figure 2. Kappa opioid receptor (KOR) signaling is linked to the Hedgehog pathway in articular chondrocytes both *in vivo* and *in vitro* (A-C).

(A) Western blot (WB) evaluation of GLI1 and pCREB. Dose dependent assay of selective KOR agonist JT09 in pig chondrocytes for 24 hours with WB quantification of pCREB and GLI1. **(B)** Evaluation of pCREB and GLI1 levels in human normal, OA, and OA treated with JT09 (10 μM) chondrocytes. **(C)** Pig articular chondrocytes treated with JT09 (10 μM) and PKA inhibitor (H89 hydrochloride 400nM). **(D)** Quantitative PCR (qPCR) of MMP13 and ADAMTS5 in OA human chondrocytes treated with JT09 (10 μM) for 24 hours. qPCR run using technical triplicates with three biological replicates. All blots are representative images (n=3). Graphs indicate Mean $\pm$ SD. P-values calculated using one-way ANOVA with post hoc Tukey test (B) or Student's *t* test (D).

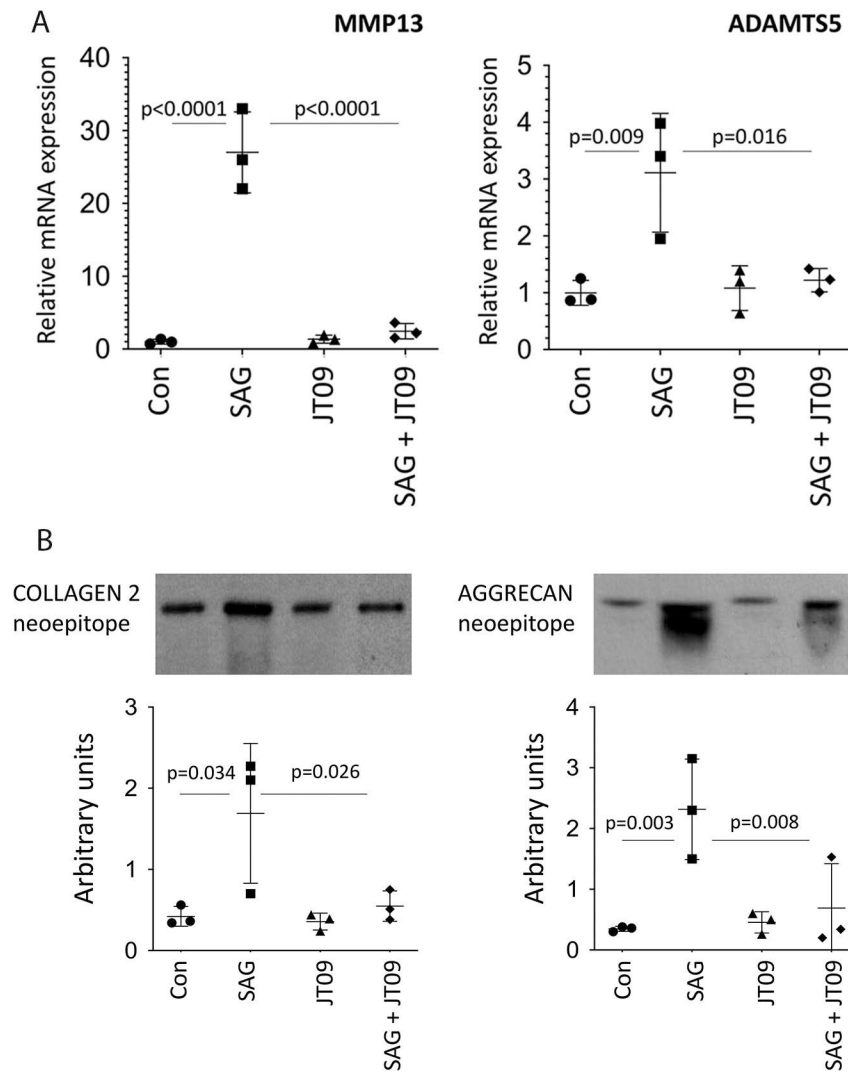


Figure 3. Hedgehog agonist SAG induces degenerative changes in cartilage cells and matrix that can be directly inhibited by an opioid agonist JT09.

(A) Transcription of catabolic genes was determined via qPCR in adult healthy pig articular chondrocytes treated with SAG (3 nM), JT09 (10 μ M), or both. qPCR run using technical triplicates with three biological replicates. Data are represented at Mean $\pm$ SD. **(B)** Adult pig articular cartilage explants were incubated with SAG, JT09, or both; levels of cleaved aggrecan and collagen epitopes in the supernatant are normalized to the wet weight of the explant. For all panels, data from three biological replicates are shown. Data are represented at Mean $\pm$ SD. P-values calculated using one-way ANOVA with post hoc Tukey test.

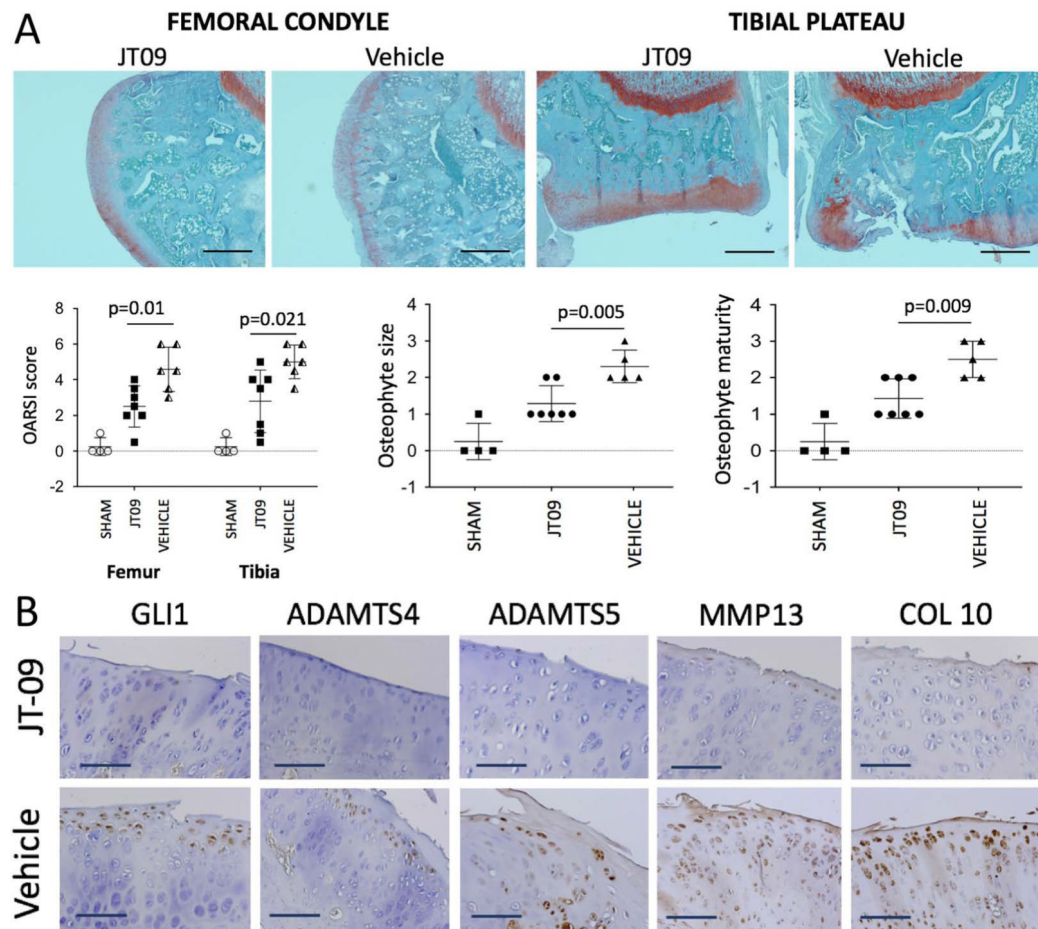


Figure 4. Selective KOR agonist JT09 attenuates cartilage degeneration in a rat model of posttraumatic osteoarthritis.

(A, B) 3 sham and 13 PMM rat joints 6 weeks after initial surgery. PMM rats received weekly intra-articular injections of saline or JT09 (300 nM). (A) Cross sectional Safranin O and Fast Green staining of tibia plateau and femoral condyle. Blinded OARSI scoring results, and measurement of osteophyte size and maturity for sham, vehicle, and JT09 treated animals. Mean $\pm$ SD (n=3, 6, and 7, respectively). P-values calculated using one-way ANOVA with post hoc Tukey test. Scale bar: 500 μ m. (B) GLI1, ADAMTS4, ADAMTS5, MMP13, and COL10 cross-sectional staining of the articular surface. Representative images (n=6, 7). Scale bar: 100 μ m. (A, B) All images taken at 10x magnification.

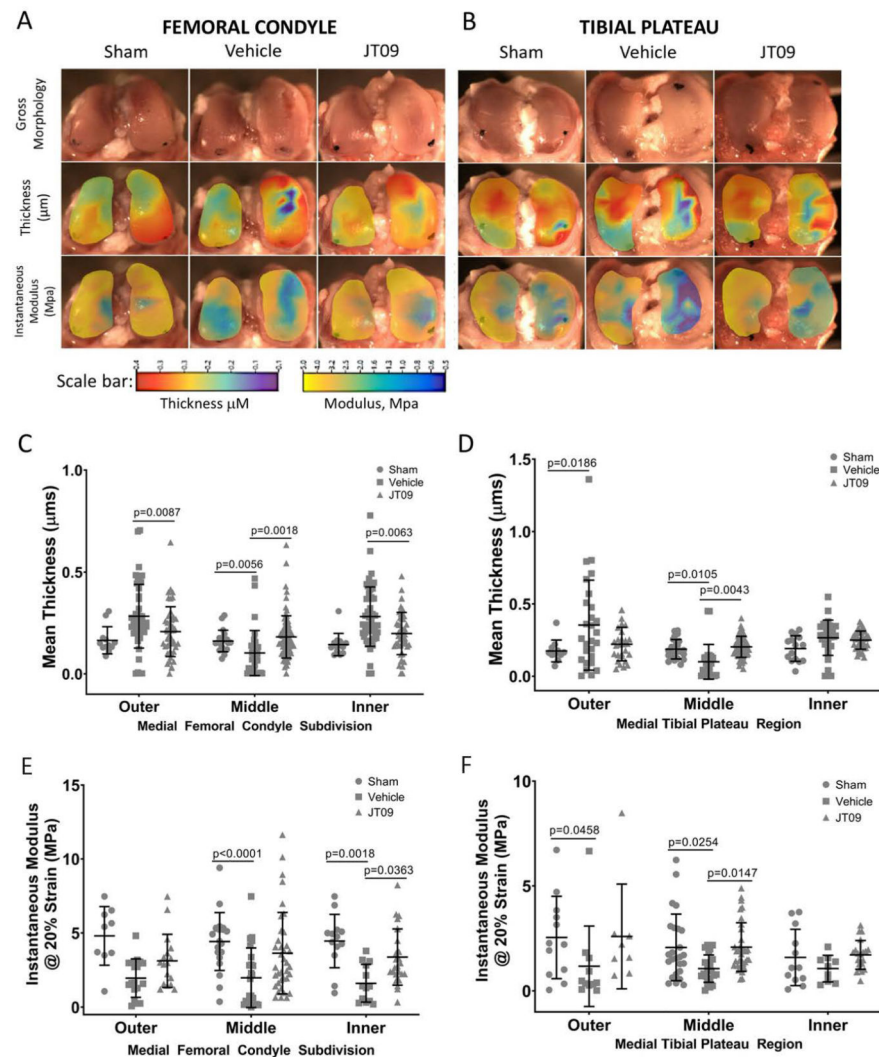


Figure 5. JT09 improves biomechanical parameters in rats after posttraumatic joint injury. (A, B) Femoral condyle (A) and tibial plateau (B) of rats treated with saline or JT09 (300 nM). Lateral (left) and medial (right) condyle of tibia plateau. Gross morphology (first row), thickness (second row) and instantaneous modulus (third row) with heat map overlays. (C-F) Thickness (C, D) and stiffness (E, F) quantification by MACH-1 indentation for sham, control, and JT09 treated animals. 5–6 data points were collected per region. Mean $\pm$ SD. P-values calculated using mixed-effects model analysis with post hoc Tukey test. Representative images (n=3, 6, 7).